

Bluetooth, and it has been released under GNU GPL v3. The Amarino toolkit consists of three main components:

- An Android application called Amarino
- The Arduino library called MeetAndroid
- The Amarino plug-in bundle (optional)

You can download these toolkit components from <http://code.google.com/p/amarino/downloads/list>. Moving on, if you want to work with the Amarino, you need to have an Android-powered device running version 2.x, though it supports version 1.6 too. Moving on to the Arduino board, you can have a Lilypad or Duemilanove, with a Bluetooth shield such as BlueSMiRF Gold and Bluetooth Mate, or an Arduino BT, which comes with Bluetooth attached to it already.

Installing Amarino is simple. After you've downloaded the *MeetAndroid* library, extract and copy it to the Arduino libraries directory. Install the *Amarino.apk* package to your Android device by downloading directly to it, or by calling `adb install Amarino.apk`. Make sure your device is connected to the computer via USB, and the PATH is set correctly for the Android SDK Tools directory.

Let's now follow the steps to get the Arduino board connected to Android:

## 1. Authentication

Open the installed Amarino application and click *Add BT Device* to search for your Arduino BT device. Make sure that it is turned ON. But before the two can talk to each other, they must be authenticated. Select the device and confirm pairing with it from the notification bar. Typically, the pin number is 1234, 12345 or 0000. Once your Arduino BT device is authenticated, it's ready to go. See Figure 2.

## 2. Creating events

The next thing is to install the plug-in bundle from <http://code.google.com/p/amarino/downloads/list> (*AmarinoPluginBundle.apk*) after which you can head to event creation. Start the Amarino application and launch the *Event Manager* of your Arduino BT device. Click *Add Event > Test Event*. This is a test event—a demo that sends a random number (0-255) every 3 seconds. Now your Android device is ready to communicate with the Arduino board, so let's set up the latter.

## 3. Setting up Arduino

Open your Arduino software and select *File > Examples > MeetAndroid > Test*. When the project opens, make a small edit: change the baud rate of your Bluetooth module from 57600 to 9600, as highlighted in Figure 3. Upload the sketch to the board. If there isn't an LED on the board, you can fix one over pin 13.

## Running the test program

Now, your Android device and Arduino board are ready to talk to each other. In the Amarino application on your Android device,



Figure 1: The Arduino LED is blinking



Figure 2: Amarino app homescreen



Figure 3: Arduino 'test' program code

click 'Connect' to communicate with the Arduino board. As soon as Android connects to the Arduino, it starts sending a random number every 3 seconds, and that lights the LED for 1 second. You can monitor the process by pressing 'Monitoring' on the main screen of the application (see Figure 4). Congratulations! You've made Android communicate with the Arduino board.

## Applications

This concept of connecting Android to Arduino can be very handy in making electronics projects more useful, by increasing their flexibility. They could easily be used in home automation controls, power consumption meters, Bluetooth-controlled robots, managing devices from the computer and much more. There is little doubt that Amarino brings us more power by helping us connect Arduino with Android. **END**

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